Signals and Systems

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$

1. *This question caries 40% of the marks.*

a) For each of the following continuous-time signals, state whether they are causal or noncausal, periodic or nonperiodic, and even, odd or non-symmetric. Give justifications for your statements. If periodic, state the period.

i) $x(t) = 1 + |\cos(\pi t)|$ [2]

Solution:
noncausal, periodic with period 1 s, even symmetric

ii) $x(t) = 2\sin^2(2t - \pi/2)$ [2]

Solution:
noncausal, periodic with period $\pi/2$, even symmetric

iii) $x(t) = \begin{cases} e^{-2t/3} & t \geq -1 \\ 0 & \text{otherwise.} \end{cases}$ [2]

Solution:
noncausal, nonperiodic, nonsymmetric

b) Consider the signal

$$x(t) = \begin{cases} t-1 & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Describe the sequence of steps that must be taken to form $y(t)$ from $x(t)$. Show the sequence of steps in illustrative sketches.

i) $y(t) = x(t + 3/2)$ [3]

ii) $y(t) = x(-2t + 1)$ [3]

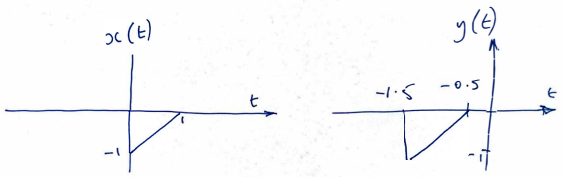
Solution:

(i)

$$y(t) = x(t+3/2) = \begin{cases} t+\frac{3}{2}-1 = t+\frac{1}{2} & -\frac{3}{2} \le t \le -\frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:

- $t \leftarrow (t+3/2)$: shift left by 1.5



(ii)

$$y(t) = x(-2t+1) = \begin{cases} [-2t+1]-1 = -2t & 0 \le t \le \frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:

- $t \leftarrow t+1$: shift left by 1
- $t \leftarrow 2t$: compress in time by factor of 2
- $t \leftarrow$: reflection in $t = 0$

[*1 mark for expression, 1 mark for description, 1 mark for sketch*]

c) An LTI system has input signal $x(t)$ and output signal $y(t)$, with Laplace transforms $X(s)$ and $Y(s)$ respectively. The system has the transfer function

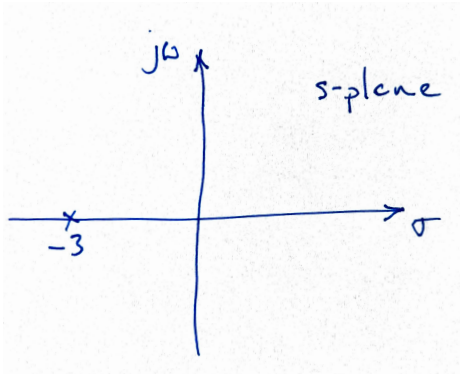
$$\frac{Y(s)}{X(s)} = \frac{3}{s+3}.$$

i) Draw a labelled sketch describing any poles and zeros of this system. [2]

ii) Write down an expression relating $y(t)$ and $x(t)$. [2]

iii) Find the impulse response of this system, assuming it is stable. [2]

Solution:
(i) The sketch should show a single pole at $s = -3$ on the left-hand half plane of s. The s-plane diagram must be fully labelled for full marks.



(ii)

$$3y(t) + \frac{dy(t)}{dx(t)} - 3x(t) = 0$$

(iii) For a stable system, the ROC includes the $j\omega$ axis, and so the ROC lies to the right of the pole. Therefore the impulse response is right-sided and given by

$$h(t) = 3e^{-3t}u(t).$$

d) Consider an LTI system with system function $H(s)$. Assuming the Region of Convergence of $H(s)$ exists, state whether the Region of Convergence of $H(s)$ is (choose any that are true): [3]

A. a region with edges parallel to the vertical axis

B. a region with edges joining the points at different frequencies using spline interpolation

C. a region with edges forming a circle around the origin

D. a region not containing any roots of the system function

E. a region never containing the origin

Solution:
A

e) i) Write down a general expression in the z-transform domain for a Finite Impulse Response filter of order M. [3]

ii) Design a Finite Impulse Response notch filter of order 2 for which the notch is at a frequency of 2.5 kHz. The filter operates at a sampling frequency of 10 kHz. [4]

Solution:
(i) The transfer function for FIR filters is

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{M} b_k z^{-k}$$

(ii) Given the sampling frequency, the notch frequency has a normalized angular frequency of $\pi/2$. Next consider 2 zeros on the unit circle at unit radius and angles $\pm\pi/2$ giving

$$H(z) = (z-j)(z+j) = z^2 + 0z + 1$$

so that the coefficients of the filter are $b = [1,\ 0,\ 1]$.

f) Consider the signal $x_1[n]$ shown in Fig. 1.1. The magnitude spectrum, $|X_1[k]|$, is computed using a 12-point DFT and is shown in Fig. 1.2. The sampling frequency used in this question is 48 kHz.

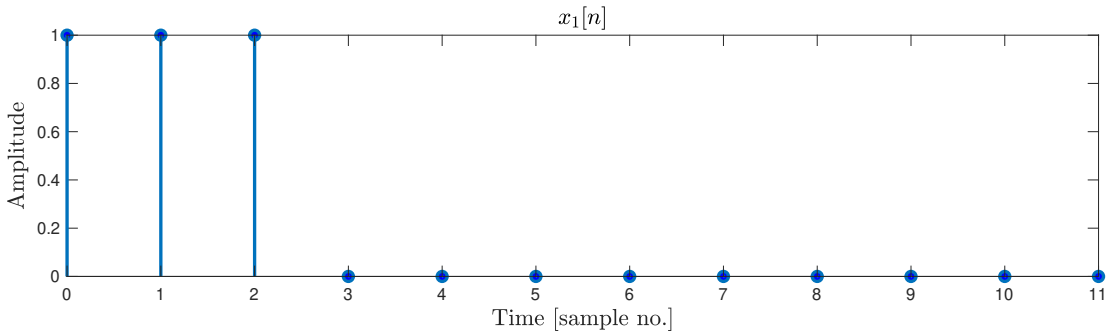


Figure 1.1

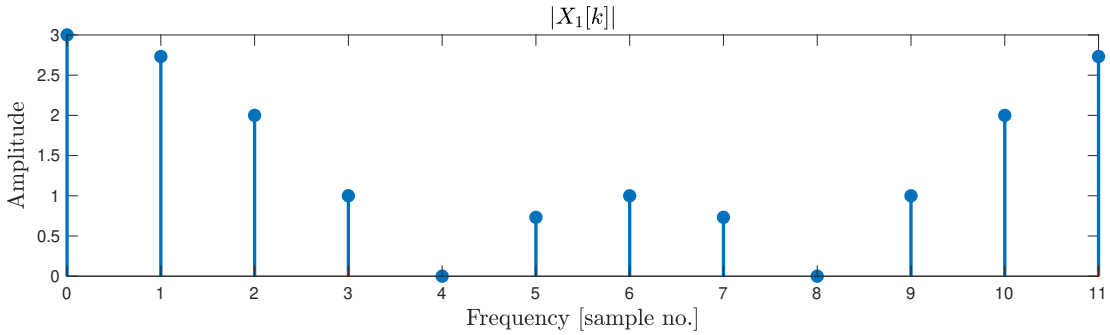


Figure 1.2

i) What is the frequency resolution, Δf of the 12-point DFT? [1]

Solution: $\Delta f = \frac{f_s}{N} = \frac{48000}{12} = 4000$ Hz

ii) Is the frequency domain representation of the signal $X_1[k]$ purely real, purely imaginary, or complex? What can you conclude about the phase of $X_1[k]$? [3]

Solution: Complex - no even/odd symmetry in the time-domain signal (2 marks). Phase is non-zero (1 mark).

iii) Now consider the signal $x_2[n]$ plotted in Fig. 1.3. The 12-point DFT is taken to obtain $X_2[k]$. Describe similarities and differences between $|X_2[k]|$ and $|X_1[k]|$, stating any property of the Fourier Transform supporting your argument. [4]

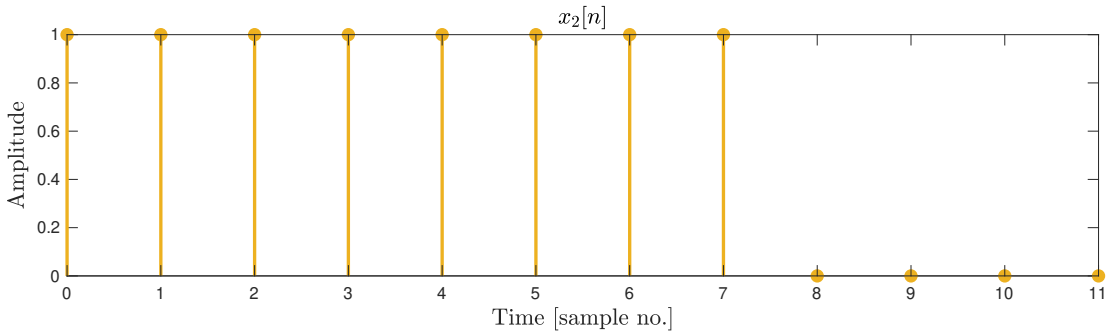
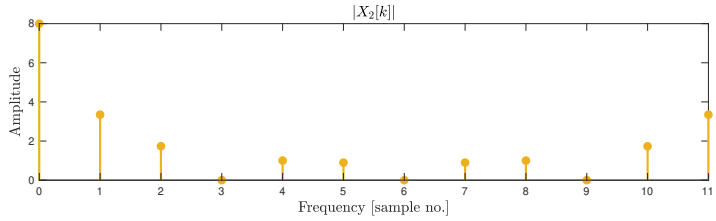


Figure 1.3

Solution: $|X_2[k]|$ is given by the plot below. The answer should include
a. Both $X_1[k]$ and $X_2[k]$ are sinc functions since both $x_1[n]$ and $x_2[n]$ are rectangular functions. (2 mark)
b. The energy in $|X_2[k]|$ is concentrated around low frequencies compared to $|X_1[k]|$ (i.e. $|X_2[k]|$ is "compressed" in frequencies) (1 mark). This is because $x_2[n]$ is a stretched version of $x_1[n]$ in the time domain. (1 mark)
c. Must include "Time scaling property of DFT" (1 mark).



iv) You now would like to remove high-frequency components in $x_1[n]$. Someone suggested setting $X_1[k] = 0$ for $k = k_c \ldots 11$, where k_c is the index of the desired cut-off frequency, and then taking the inverse DFT to obtain $x_{lp}[n]$. What is wrong with this approach? Suggest a more appropriate method. [4]

Solution: The suggested approach breaks the conjugate-symmetry of the DFT (1 mark). Moreover, setting frequency bins directly to zero leads to a sinc interpolation and does not ensure the desired response in between DFT sample points (2 marks). A better method would apply time-domain filtering using FIR or IIR filters (1 mark)

2. *This question caries 30% of the marks.*

a) Consider the classes of continuous-time signals and discrete-time signals. For each, further consider the classes of periodic signals and nonperiodic signals.

State the Fourier representation appropriate for each class of signal and give the corresponding formulae to obtain the Fourier representation from the time domain signal. [4]

Solution:

	Periodic	**Nonperiodic**
Continuous-Time (t)	Fourier Series $x(t) = \sum_k A[k] e^{jk\omega_0 t}$	Fourier Transform $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$
Discrete-Time $[n]$	Discrete-Time Fourier Series $x[n] = \sum_k A[k] e^{jk\Omega_0 n}$	Discrete-Time Fourier Transform $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$

b) One period of a particular periodic signal with period $T = 3$ s is defined as

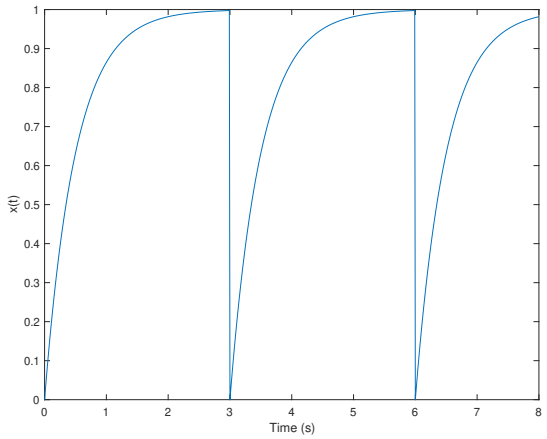
$$x(t) = 1 - e^{-2t} \quad 0 \le t \le T.$$

i) Sketch $x(t)$ for $0 \le t \le 8$ s. [3]

ii) Derive an appropriate Fourier representation of $x(t)$. [3]

iii) Hence determine the magnitude spectrum of the signal at $1/3$ Hz. [3]

Solution:
(i)



(ii) Since the signal is periodic, use the continuous-time Fourier series.

$$\begin{aligned} X[k] &= \frac{1}{3}\int_0^3 \left(1-e^{-2t}\right)e^{-jk2\pi t/3}dt \\ &= \frac{-1}{jk2\pi}\left[e^{-jk2\pi}-1\right] - \frac{-1}{6+jk2\pi}\left[e^{(-6+jk2\pi)}-1\right] \end{aligned}$$

(iii) The frequency of $1/3$ Hz corresponds to the fundamental frequency, meaning $k=1$, so that

$$\begin{aligned} X[1] &= \frac{1}{6+jk2\pi}\left[e^{-6}-1\right] \\ |X[1]| &= \frac{e^{-6}-1}{\sqrt{6^2+4\pi^2}}. \end{aligned}$$

c) The frequency response of a particular continuous-time system is

$$H(j\omega) = \frac{10}{j\omega+10}$$

and the input signal is $x(t)=\cos(4\pi t)$.

i) Write down the Fourier representation of the input signal.[3]
Hint: consider using terms of the form $\delta(\omega+\ldots)$.

ii) When $x(t)$ is used as the input signal, find the Fourier representation of the output signal, and write it in the form $A(R+jI)$ where A is a scalar, and R and I are the real and imaginary terms respectively. [3]

iii) Hence find the time domain output signal, $y(t)$. [2]

Solution:
(i)

$$X(j\omega) = \pi\left(\delta(\omega-4\pi)+\delta(\omega+4\pi)\right)$$

(ii)

$$Y(j\omega) = \pi\left(\delta(\omega-4\pi)+\delta(\omega+4\pi)\right)\frac{10}{j\omega+10}$$

$$= \frac{10\pi}{16\pi^2+100}\left[10\Big(\delta(\omega+4\pi)+\delta(\omega+4\pi)\Big)+j4\pi\Big(\delta(\omega+4\pi)-\delta(\omega-4\pi)\Big)\right]$$

(iii)

$$y(t) = 0.388\cos(4\pi t)+0.487\sin(4\pi t).$$

d) Consider a continuous-time LTI system characterized by the system function $H(s)$. When the input signal to the system is $x(t)=\delta(t)$, the corresponding output signal is

$$y(t) = \frac{2}{3}e^{3t}u(-t)+\frac{2}{3}e^{-t}u(t).$$

i) Find $H(s)$ and state the Region of Convergence. [4]

ii) Sketch a representation of $H(s)$ on the s-plane and indicate the Region of Convergence. [2]

iii) The input signal is now changed to the unit step function so that $x(t)=u(t)$. Find the Laplace transform of the corresponding output signal and state its Region of Convergence. [3]

Solution:
(i)

$$H(s) = \frac{Y(s)}{X(s)} = \frac{4}{3}\frac{(s-1)}{(s-3)(s+1)}$$

ROC is given by $-1<\Re\{s\}<3$.
(ii) The sketch must show a zero at $s=1$ and poles at $s=-1$, $s=3$. ROC must be bounded between vertical lines at $\sigma=-1$ and $\sigma=3$.
(iii)

$$X(s) = \frac{1}{s} \quad \text{ROC: } \Re\{s\}>0$$

The output signal LT is given by

$$Y(s) = H(s)X(s) = \frac{4}{3}\frac{(s-1)}{s(s-3)(s+1)} \quad 0<\Re\{s\}<3.$$

3. *This question caries 30% of the marks.*

a) The discrete-time signal $x[n]$ is defined as

$$x[n] = \begin{cases} 1 & 0 \le n < 5 \\ 0 & \text{otherwise.} \end{cases}$$

i) Find the DTFT of $x[n]$ and express the DTFT of $x[n]$ as a ratio of sine functions. [8]

ii) Sketch the magnitude and phase of the DTFT of $x[n]$. [3]

Solution:

$$X\left(e^{j\Omega}\right) = \sum_{0}^{4} 1.e^{-j\Omega n}$$

Using

$$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r} \quad \text{with } N = 5$$

leads to

$$X\left(e^{j\Omega}\right) = \frac{1-e^{-j\Omega N}}{1-e^{-j\Omega}}$$

[3]

To express in terms of ratio of sine functions,

$$\begin{aligned} X\left(e^{j\Omega}\right) &= \frac{e^{-j\Omega N/2}\left(e^{j\Omega N/2} - e^{-j\Omega N/2}\right)}{e^{-j\Omega/2}\left(e^{j\Omega/2} - e^{-j\Omega/2}\right)} \\ &= e^{-j\Omega(N-1)/2}\frac{\sin(\Omega N/2)}{\sin(\Omega/2)}. \end{aligned}$$

[3]

For $\Omega = 0, \pm 2\pi, \pm 4\pi \ldots$

$$X\left(e^{j\Omega}\right) = N.$$

[2]

The sketch of the magnitude must show (i) the general shape of the sinc function, (ii) periodicity of 2π. The sketch of the phase must show (iii) a tilt in the negative gradient direction associated with the linear phase shift of $(N-1)/2$. 1 mark for each. [3]

b) The sound of a large bell known as Big Ben is to be recorded using a microphone and a digital sound recording device with variable input gain.

i) Describe how you would determine the minimum sampling frequency fs Samples/s that should be used for the recording, including a brief description of any measurements or parameters that would be needed to detemine fs. [3]

ii) Assume a recording has been made at an appropriate sampling frequency fs Samples/s, and a histogram of the samples has been plotted as represented in Fig. 3.1.

Explain the peak in the histogram at zero amplitude. [1]

Explain the peaks in the histogram at amplitudes of ± 1000. State whether these peaks are an expected characteristic of this recording and suggest a reason they occur in the histogram. [4]

Explain what could be done so that the peaks at amplitudes of ± 1000 would not occur in the histogram. [2]

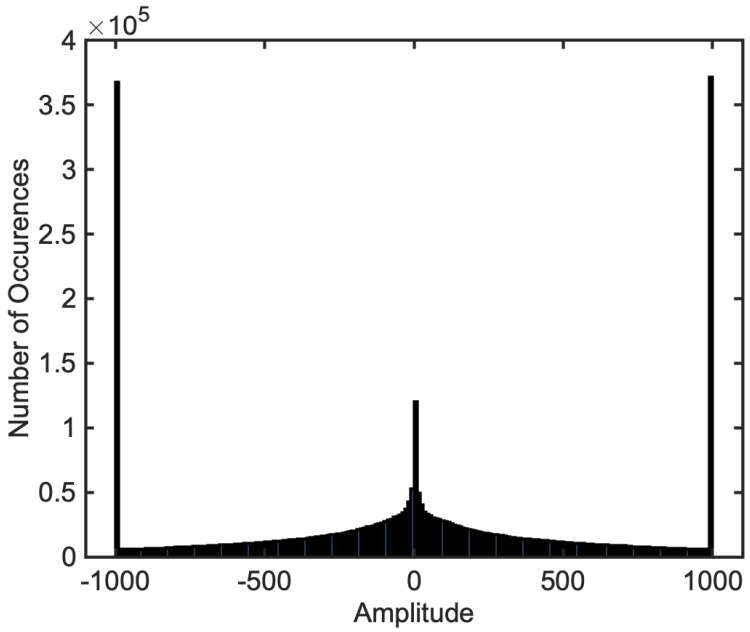


Figure 3.1

Solution:
(i)
Option a): use a spectrum analyzer to inspect the sound spectrum of the bell and determine the maximum frequency containing significant energy; choose $fs > 2\times$ the maximum frequency.
Option b): In an experiment, gradually increase the sampling frequency used to record the bell until no significant additional energy is captured when the sampling frequency is increased.
(ii)
Peak at zero is due to pause/silence before the bell sound.
Peaks at 1000 are due to amplitude clipping. Not expected.
Reducing the input gain of the recording device will better accommodate the dynamic range of the bell sound.

c) Consider a discrete-time LTI system with impulse response $h[n]$ and system function $H(z)$ with

$$h[n] = [1,\ -0.1,\ 0.12,\ -0.04].$$

i) Write an expression for $H(z)$. [1]

ii) Find the magnitudes of $H(z)$ at normalized angular frequencies of 0 and π, and express these in decibels. [3]

Solution:
(i)

$$H(z) = 1 - 0.1z^{-1} + 0.12z^{-2} - 0.04z^{-3}$$

(ii)

$$|H(z)|_{(z=1)} = 0.98 = -0.18 \text{ dB}$$
$$|H(z)|_{(z=-1)} = 1.26 = 2 \text{ dB}$$

d) Consider an LTI system described in the z-transform domain by the system function

$$H(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})} \quad \text{for } |z| > \frac{1}{3}.$$

Find the impulse response of the system, $h[n]$. [5]
Hint: consider using partial fraction expansion

Solution:

$$H(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}$$

Considering the ROC as stated, the ROCs of the individual terms can be seen to be outside both the corresponding poles leading to

$$h[n] = \left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{3}\right)^n u[n].$$